With the Pandas UDF set up, we read our Delta tables with images from our volumes and score them with the `apply_cv` function.

```
spark.conf.set("spark.sql.execution.arrow.maxRecordsPerBatch", 32)
predictions_df = spark.read\
  .format("delta")\
  .load(f"/Volumes/{catalog}/{database_name}/files/intel_image_clf/valid_imgs_main.delta")\
  .withColumn("prediction", apply_cv("content"))
```

Figure 6.36 – Scoring all images in our volume

Once your scores are done, you can save them to another table for further investigation. With that, we have trained and scored our Multilabel classification model. We will add some infrastructure around the model in *Chapter 7* to make it production-ready.

Summary

We have now turned all our hard work from previous chapters into predictive models. From building training sets with the Databricks Feature Engineering client, to baselining models with AutoML and tracking experiments in MLflow, we covered core stages of all data science projects. We also discussed the Databricks Marketplace and Databricks LLM Playground, two tools that can further enhance data science and machine learning tasks by incorporating external data and models with your existing work.

You now have the skills required to take full advantage of the data available to you and gain powerful, predictive insights from the models you train. Of course, building the model is just the start – in *Chapter 7, Productionizing ML* on Databricks, we show you how to get your models in the hands of your users.

Questions

The following questions solidify key points to remember and tie the content back to your experience.

1. Why is it important that MLflow is open-source?
2. Why would you use a baseline model? Which Databricks Platform tool makes it easy to experiment and build baseline models rapidly?
3. What are some tools you can use to incorporate external data and models into your own data science projects? When might external data artifacts be useful?